Abdelrahman Alnaggar

Data Analyst

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Summary

Results-driven Data Analyst & Business Intelligence Specialist with a strong foundation in Business Information Systems and end-to-end data pipelines. Proficient in leveraging SQL, Python, and advanced Excel (Power Query, Power Pivot) to extract, transform, and model large-scale datasets into actionable commercial strategies. Proven track record of designing executive Power BI dashboards with complex DAX measures and deploying predictive machine learning models that optimize core KPIs, streamline reporting efficiency, and drive data-backed decision-making across diverse business operations.

Education

- **Egyptian Institute of Alexandria Academy For Management and Accounting** Sep 2023 – Jun 2027
  Bachelor of Business Information Systems (BIS)
  Grade: Excellent (92%)

Experience

Data Analyst Specialist Program | Digital Egypt Pioneers Initiative (DEPI) Nov 2025 – Present

- Engineered data preparation and cleaning pipelines using Python and SQL, improving data quality by 15% and reducing processing anomalies by 30%.
- Designed and deployed 7 interactive dashboards and reports using Tableau and Power BI, enabling 12+ stakeholders to make data-driven decisions.
- Implemented data governance standards to maintain data quality, enhance security protocols, and ensure 100% regulatory compliance.

Artificial Intelligence for Business | National Telecommunication Institute (NTI) Aug 2025 – Sep 2025

- Architected and deployed 3 supervised machine learning models utilizing Python, Scikit-learn, and TensorFlow to address key business objectives.
- Applied robust data preprocessing, feature engineering, and model evaluation techniques in 5 end-to-end predictive pipelines.
- Engineered a predictive flight ticket pricing model utilizing Random Forest regression, achieving an R2 score of 0.85 and a 1.5% mean absolute percentage error (MAPE).

AICE Program | ALX Africa Sep 2024 – Nov 2024

- Integrated AI productivity tools into daily workflows to optimize efficiency, saving 5+ hours weekly on task automation and complex troubleshooting.
- Collaborated with cross-functional teams of 5 members to execute project milestones, honing professional communication, teamwork, delivering technical demonstrations and aligning project objectives.
- Formulated and delivered 3 professional presentations, demonstrating technical solutions to stakeholder audiences with clarity and impact.

Projects

Heart Disease End-to-End Prediction & Analysis: **[View Project]**

- Engineered an end-to-end ML pipeline in Python for heart disease risk prediction, training and evaluating 6 models (Logistic Regression, Random Forest, XGBoost, LightGBM, Neural Network, Stacking Ensemble) on 319K+ CDC patient records.
- Designed a complete Medallion Data Warehouse Architecture (Staging → Bronze → Silver → Gold) in SQL Server with an automated Python ETL pipeline and pipeline observability logging.
- Optimized model performance using Bayesian hyperparameter tuning via Optuna (200 trials), achieving a best ROC-AUC of 0.84 and 80.57% Recall with LightGBM.
- Deployed a real-time Streamlit web app with live SHAP explainability and built an interactive Power BI dashboard featuring K-Means patient segmentation and risk analysis.

NexVault – Financial Intelligence Dashboard Suite: **[View Project]**

- Architected a comprehensive financial intelligence analytics suite utilizing Power BI across four specialized dashboards to consolidate transaction and compliance data.
- Visualized and analyzed macro business metrics, monitoring over $4.26B in total revenue, $1.66B in profit, and achieving a 64.11% compounding ROI to enhance executive oversight.
- Segmented risk and compliance metrics, analyzing a 77.28% KYC verification rate and mapping $219.66M in fraud losses to support the refinement of AI fraud-detection models.

RetailNova Sales Dashboard: [View Project]

- Built an interactive sales monitoring dashboard in Excel to track retail performance across 4 regions and 8 product categories from 2022 to 2025.
- Processed and analyzed transaction-level sales data containing over 31K total orders, tracking $109.8M in revenue and $9.3M in profit.
- Extracted actionable insights on product performance and sales representative efficiency, identifying key revenue drivers (such as Bikes contributing 86% of total revenue) to guide inventory and sales strategy.

Skills

- **Data Analysis & Visualization:** Excel, Power BI, Tableau, SQL, Pivot Tables, Power Pivot, Power Query
- **Programming & Databases:** Python (Pandas, NumPy, Matplotlib), SQL (Microsoft SQL Server, Queries, Data Warehousing)
- **Machine Learning & Statistics:** Supervised Machine Learning, Statistical Analysis
- **Soft Skills:** Problem-Solving, Teamwork, Attention to Detail, Time Management, Communication Skills

Certification

- **Data Analyst Specialist – DEPI (Digital Egypt Pioneers Program)**
- **Data Analysis – Microsoft**
- **Artificial Intelligence for Business – NTI (National Telecommunication Institute)**
- **Deep Learning Fundamentals – NVIDIA**
- **ALX AICE – AI Career Essentials**